
Half-period symmetries for sequences modulo 3

Mathematical formalization and algorithmic audit

This note rigorously separates three phenomena that are often conflated: cycle negation, half-period antiperiodicity, and reversal. It proves the unique opposite shift of a nonzero primitive word, gives combinatorial, spectral, and matrix characterizations, and connects these results to an exhaustive analyzer of modular recurrences.

Repository: Modular-recurrence-symetry-analyzer-
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1. Purpose

This note formalizes the phenomena described in the repository [59200/k-bonacci-pisano-finder](#) under the informal phrases “periods complementary to three” and “self-complementary when cut in two”.

The invariant setting is that of periodic words over the field

$$\mathbb{F}_3 = \{0, 1, 2\},$$

with additive involution

$$\nu(x) = -x \pmod{3}, \quad \nu(0) = 0, \quad \nu(1) = 2, \quad \nu(2) = 1.$$

The phrase “complement to 3” should therefore be replaced by **additive inverse modulo 3** or **complement by negation modulo 3**. The decimal sum of the chosen representatives is not always 3, because 0 is mapped to 0.

2. Three distinct properties

Let $w = (w_0, \dots, w_{T-1}) \in \mathbb{F}_3^T$ be a word of primitive period T .

2.1 Pair of opposite periods

Two periods w and v form an opposite pair if

$$v_n = -w_n \pmod{3}$$

for every n . They may belong to two distinct cycles.

2.2 Half-period antiperiodicity

The word w is **half-period antiperiodic** if $T = 2h$ and

$$w_{n+h} = -w_n \pmod{3}$$

for every n . It can then be written

$$w = H \parallel (-H), \quad H = (w_0, \dots, w_{h-1}).$$

Examples:

$$01120221 = 0112 \parallel 0221, \quad 011022 = 011 \parallel 022, \quad 01220211 = 0122 \parallel 0211.$$

2.3 Reversal of the second half

The operation

$$\mathcal{R}(a_0, \dots, a_{h-1}) = (a_{h-1}, \dots, a_0)$$

is independent of negation. In the Fibonacci example,

$$\mathcal{R}(0221) = 1220.$$

Thus, 1220 is the **reversed antipodal word** of 0112, not the second half itself.

3. The unique opposite-shift theorem

Theorem. Let w be a nonzero primitive word of period T . If there exists a shift r such that

$$w_{n+r} = -w_n$$

for every n , then T is even and

$$r \equiv \frac{T}{2} \pmod{T}.$$

Proof

Applying the shift twice gives $w_{n+2r} = w_n$. Primitivity implies $T \mid 2r$. The shift r is not zero modulo T , because a nonzero word over $\mathbb{F}_3$ cannot satisfy $w = -w$. The unique nontrivial element of order 2 in the cyclic group $\mathbb{Z}/T\mathbb{Z}$ exists only when T is even and equals $T/2$. $\square$

Consequence

A nonzero primitive period is therefore either:

1. sent by negation to a distinct cycle;
2. invariant under negation, in which case it is necessarily half-period antiperiodic.

This dichotomy removes the confusion between the pair

$$00111201, \quad 00222102$$

and intrinsically antiperiodic periods such as 011022.

4. Combinatorial consequences

If $w = H \parallel (-H)$, then

$$\#\{n : w_n = 1\} = \#\{n : w_n = 2\}, \quad \#\{n : w_n = 0\} \equiv 0 \pmod{2},$$

and

$$\sum_{n=0}^{T-1} w_n = 0 \quad \text{in } \mathbb{F}_3.$$

These conditions are necessary but not sufficient. Using the centered lift

$$\tilde{0} = 0, \quad \tilde{1} = 1, \quad \tilde{2} = -1,$$

the generating polynomial factors as

$$W(X) = \sum_{n=0}^{2h-1} \tilde{w}_n X^n = (1 - X^h) \sum_{n=0}^{h-1} \tilde{w}_n X^n.$$

5. Fourier characterization

Let

$$\widehat{w}(k) = \sum_{n=0}^{2h-1} \tilde{w}_n e^{-2\pi i k n / (2h)}.$$

Then $w_{n+h} = -w_n$ if and only if $\widehat{w}(k) = 0$ for every even index k . When $w_{n+h} = -w_n$,

$$\widehat{w}(k) = \left(1 - (-1)^k\right) \sum_{n=0}^{h-1} \tilde{w}_n e^{-2\pi i k n / (2h)},$$

so the factor vanishes for even k . Conversely, if every even mode vanishes, the word lies in the subspace spanned by odd modes. A shift by h acts on mode k by $(-1)^k = -1$, hence by $-I$ on the word. This is an exact spectral diagnostic, not merely a visual comparison of the two halves.

6. Linear sequences modulo 3

Consider an order- k recurrence

$$u_{n+k} = c_0 u_n + c_1 u_{n+1} + \cdots + c_{k-1} u_{n+k-1} \pmod{3}.$$

Its state is $s_n = (u_n, \dots, u_{n+k-1})^\top$, and its evolution is $s_{n+1} = M s_n$, where M is the companion matrix.

6.1 Pure periodicity

The determinant of M equals c_0 up to sign. Over $\mathbb{F}_3$,

$$M \text{ is invertible} \iff c_0 \neq 0.$$

In that case every state orbit is purely periodic. If $c_0 = 0$, a preperiod may occur and must be separated from the cycle.

6.2 Negation of cycles

Linearity gives $M(-s) = -M(s)$. Negation therefore sends each cycle to a cycle of the same length. For a nonzero cycle, only two cases are possible:

- two distinct cycles exchanged by negation;
- one invariant cycle, necessarily half-period antiperiodic.

The zero cycle is the unique degenerate exception: it is fixed pointwise by negation and does not define a nontrivial primitive antiperiod.

6.3 Criterion for one seed

For a seed s_0 , the relation

$$M^h s_0 = -s_0$$

implies, for every linear output of the state,

$$u_{n+h} = -u_n.$$

When $s_0 \neq 0$ and the cycle is primitive, this relation yields the half-period antiperiod described above.

6.4 Uniform criterion

All seeds satisfy the same antiperiod relation h if and only if

$$M^h = -I.$$

Equivalently, the minimal polynomial μ_M satisfies

$$\mu_M(X) \mid X^h + 1 \quad \text{in } \mathbb{F}_3[X],$$

and therefore $M^{2h} = I$.

7. Fibonacci example

For

$$M = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \quad \text{over } \mathbb{F}_3,$$

one computes $M^4 = -I$ and $M^8 = I$. The period generated by the seed $(0, 1)$ is

$$01120221 = 0112 \parallel (-0112).$$

Its reversed second half is $\mathcal{R}(0221) = 1220$. The numerical coincidence, under decimal reading,

$$1220 = L_{14} + F_{14} = 2F_{15}$$

is an additional encoding identity, not a general consequence of antiperiodicity.

8. Generalization to arbitrary sequences modulo 3

No recurrence assumption is needed to analyze a given periodic word. For every primitive period one may compute:

- its rotation cycle;
- its additive inverse;
- its affine symmetries $w_{n+r} = aw_n + b$;
- its reversal symmetries $w_{r-n} = aw_n + b$;
- any antiperiodicity;
- its even and odd Fourier modes.

Matrix theory is needed only when a declared linear recurrence is available.

9. Extension to a general modulus

Over $\mathbb{Z}/m\mathbb{Z}$, the natural involution remains $x \mapsto -x$. The definition

$$w_{n+h} = -w_n \pmod{m}$$

remains valid. For $m = 2$, negation is the identity and antiperiodicity collapses to ordinary periodicity. For $m > 2$, the distinction is nontrivial. The phrase “complement to m ” should not be used as an algebraic definition because it depends on the chosen integer representatives; the invariant term is “additive inverse modulo m ”.

10. Algorithmic audit

A period of an order- k recurrence modulo m must be detected on the finite state space

$$(\mathbb{Z}/m\mathbb{Z})^k,$$

not by searching for a repeated block in an arbitrarily long prefix. The supplied script uses:

- compact base- m state encoding;
- Brent's algorithm for one seed, with $O(\mu + \lambda)$ time and $O(1)$ memory;
- exact functional-graph decomposition for all seeds, with $O(m^k)$ time;
- parallelization across coefficient families;
- separate classification of opposite cycles and antiperiodic cycles;
- a matrix check of $M^h = -I$.

No Mathematica kernel or external wrapper is required.

11. Reproducible scan results

The scan of the fifteen named repository families recovers, for the tritetranacci family, distinct opposite pairs of lengths 24 and 8, together with the antiperiodic periods 011022, 01220211, and 12. A second scan covers the 119 nonzero binary coefficient vectors of orders 2 through 6:

$$\sum_{k=2}^6 (2^k - 1) = 119.$$

The script finds:

- 13 families satisfying a global criterion $M^h = -I$;
- 88 families with at least one nontrivial half-antiperiodic cycle;
- 96 families with at least one distinct opposite-cycle pair.

These are exhaustive experimental results within this finite window. They are reproducible from `mod3_symmetry_scan_summary.json`, `binary_recurrence_families_mod3.json`, and `binary_recurrence_families_mod3_summary.csv`.

Conclusion

Negation modulo 3 acts as a structural involution on periodic words and cycles of linear recurrences. For a nonzero primitive word, invariance under this involution can occur only through the unique half-period shift. The same property is recognized equivalently by factorization of the generating polynomial, vanishing of even Fourier modes, or, in the linear setting, by a matrix relation $M^h = -I$. The algorithmic analysis therefore separates intrinsic cycle symmetries from mere associations between opposite cycles.